Vivaswan Rai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.46
Aklank Public School, Basant Vihar, Kota	Class XII (CBSE)	2024	95.20%
Seth Anandram Jaipuria School, Kanpur	Class X (ICSE)	2022	97.83%

RESEARCH EXPERIENCE

Mechanistic Analysis of Catastrophic Forgetting in LLM Fine-Tuning

*Undergraduate Research Assistant/ Guide: Dr. Srijith PK,CSE,IIT Hyderabad - Github/**Manuscript** in preparation, targeting **CVPR** and **tier-two venues***

- Identified a **causal computation zone** (layers 10-22) governing multi-step reasoning via layer ablation, **Fisher Information** (6× sensitivity differential), and permutation experiments on an RTX 3090 GPU server using **lm-evaluation-harness**.
- Implemented Protected SFT (freezing critical layers, **~60% forgetting reduction**) and Adaptive Self-Rehearsal with real-time monitoring via custom HuggingFace TrainerCallback hooks, eliminating reasoning forgetting entirely (**15pp to +0.2pp**).
- Proposed a **three-criteria prediction framework** (multi-step chaining × formal operations × sequential state dependence) for identifying which capabilities degrade under SFT, validated across 13 benchmarks, 57 MMLU subjects, and 3 model families (Qwen2.5-3B, Phi-2, LLaMA 3.2-3B).

Explainable Multi-Lead ECG Classification with Contrastive Retrieval-Augmented Explanations

*Undergraduate Research Assistant/Guide: Dr Nagarajan Ganpathy***-** *Github*

- Built a **CNN-Transformer hybrid** (ECA attention, 4-stage residual CNN) with a **custom PyTorch DataLoader** (on-the-fly wfdb parsing, sliding-window segmentation, composable augmentation) for **5-class ECG classification** on PTB-XL (21K+ records, **Macro-F1=0.68**).
- Implemented contrastive IG attribution from scratch and a **FAISS retrieval pipeline** with a 4-layer verification stack, reducing LLM explanation hallucination by 60%.

PROJECTS

Implied Volatility Surface Reconstruction using Beta-VAE - *Github*

- Built a streaming **JSON data pipeline** to parse millions of options records, estimate missing spot prices via **ATM-delta heuristics**, bin to a **log-moneyness × maturity grid**, and export normalized tensor bundles.
- Designed a **coordinate-aware -VAE** (CoordConv, masked L1, KL annealing, no-arbitrage constraints); achieved **57% MAE reduction** over PCA/bicubic/Soft-Impute baselines.

Customer Churn Prediction & Retention Strategy using XGBoost - *github*

- Built an end-to-end churn prediction pipeline with engineered behavioral features (spending momentum, engagement decay, utilization ratios), achieving **96%+ ROC-AUC** and identifying top **churn drivers** via **SHAP analysis**.
- Framed model output as a business case -- estimated ~$200K+ revenue retention by targeting top predicted churners with interventions, demonstrating a 10x ROI over intervention costs.

GPT-2 Language Model -- From-Scratch Implementation - *Github*

- Implemented **124M-parameter GPT-2** in PyTorch **from scratch** (BPE tokenizer, causal multi-head self-attention, GELU, layer norm) with a custom **pretraining pipeline** (AdamW, cosine annealing, gradient clipping).

HTTPS Web Server From Scratch (C) - *Github*

- Built a **TCP server** using raw **POSIX sockets** and a custom **HTTP/1.1 parser** (method/header extraction via pointer arithmetic, content-type routing, error handling) -- no frameworks.

- Integrated **OpenSSL for TLS** termination with **full handshake support**; validated via curl, Wireshark, and gdb.

SKILLS

- **Programming Languages**: C, C++, Python, SQL
- **Frameworks and Libraries**: PyTorch, TensorFlow, Scikit-Learn, XGBoost, OpenCV, keras, HuggingFace Transformers, FAISS, OpenSSL
- **Tools & Technologies**: Git, Docker, POSIX Sockets, lm-evaluation-harness

RELEVANT COURSEWORK

Mathematics : Probability & Random variables, Discrete Maths, Complex Variables, Calculus, Introduction to Statistics, Linear Algebra

Computer Science: Data Structures, Algorithms, Operating Systems, Introduction to AI

POSITIONS OF RESPONSIBILITY

Core at Kludge (Cybersecurity and Networking Club, IIT Hyderabad)

- Created challenges for CTFs in cryptography and web security.
- took sessions on CTFs and Cryptography to mentor the students

Institute Tennis Coordinator

- **Represented** and **lead** the IIT Hyderabad Tennis team at the 58th **Inter IIT sports meet**.
- Pioneered initiatives to enhance Tennis culture on campus, encouraging broader student participation.
- Successfully managed and organised 3 major college level tennis tournaments, Oversaw end-to-end logistics, scheduling and execution thus developing strong skills in **Leadership and Management**.

Ideation Coordinator - Finance and Consulting club, IIT Hyderabad

- Ideated and designed the main events and their execution workflow for the club's flagship event, Horizon'26.
- Helped with speaker outreach and content curation, and oversaw successful completion of the Campus Heist.